

# Problems in Group Theory Chapter 1

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February 2025

## 1 Problems

Q 1. For any  $H < G$  are the left and right  $H$ -cosets in bijection with each other?

Q 2. If  $H, K < G$  and  $H \cong K$ , is it true  $[G : H] = [G : K]$ ?

T 1. If  $H < K < G$  and  $[G : H]$  is finite, then  $[G : K]$  is finite as well, and  $[G : H] = [G : K][K : H]$ .

Consider the surjective map  $\varphi : G/H \rightarrow G/K$ . Since  $G/H$  is finite,  $G/K$  is also finite.  $\ker(\varphi) = K/H$ , so  $[G : H] = [G : K][K : H]$ , as desired.

T 2. Suppose  $A, B < G$  and  $[G : B]$  is finite. Show that  $[A : A \cap B] \leq [G : B]$  and that equality holds iff  $G = AB$ .

Consider the map  $\varphi : A/B \hookrightarrow G/B$ . Since  $G/B$  is finite, so is  $A/B$ . There are  $[A : A \cap B]$  elements in  $A/B$ , so  $[A : A \cap B] \leq [G : B]$ , as desired. Suppose equality holds, so that  $G/B = A/B$ . For any  $g \in G$ ,  $gB = aB$  for some  $a \in A$ , or  $g = ab$  for some  $b \in B$ , or  $g \in AB$ . We know  $AB \subset G$ , so  $G = AB$ . For the reverse direction,  $G/B \subset A/B$  and  $A/B \subset G/B$ , and we're done.

T 3. Let  $S \subset G$ . Then the number of  $G$  conjugates of  $S$  is  $[G : N(S; G)]$ .

$G$  acts on  $\mathcal{P}(G)$  by conjugation. By Orbit-Stabilizer  $S$  has  $[G : N(S; G)]$  conjugates.

P 3. What is the order of a group  $G$  generated by elements  $x, y$  subject only to the relations  $xy^2 = y^3x$  and  $yx^3 = x^2y$ ?

P 7. Let  $H < G$  such that  $[G : H] < \infty$ . Show that there exists  $N < H$  such that  $N \triangleleft G$  and  $[G : N] < \infty$ .

Lemma. Define  $GHG^{-1} = \{gHg^{-1} : g \in G\}$ . If  $G/H$  is finite, then  $|GHG^{-1}| \leq [G : H]$ .

Proof. Consider the map  $\varphi : G/H \rightarrow GHG^{-1}$  given by  $\varphi(gH) = xHx^{-1}$  for any  $x \in gH$ . This map is well defined because  $x = gh$  for some  $h \in H$ , and  $xHx^{-1} = ghHh^{-1}g^{-1} = gHg^{-1}$ . This map is surjective, proving the claim.

Consider the subgroup  $K = \cap_{g \in G} gHg^{-1}$ . By the Lemma,  $K$  is a finite intersection. We make some observations.  $[G : gHg^{-1}] < \infty$  for all  $g$ .  $[G : K] < \infty$  by Poincare.  $K < H$ .  $K \triangleleft G$ . Hence,  $K$  is our desired subgroup.

P 8. If  $A, B < G$  and  $[G : A], [G : B]$  are finite and relatively prime, then  $G = AB$ .

By Poincare,  $[G : A \cap B]$  is finite. Then  $[G : A][A : A \cap B] = [G : A \cap B] = [G : B][B : A \cap B]$ , where  $[A : A \cap B]$  and  $[B : A \cap B]$  are finite. Rearranging,

$$\frac{[G : A]}{[G : B]} = \frac{[B : A \cap B]}{[A : A \cap B]}.$$

Since  $[G : A] \perp [G : B]$ ,  $[G : A] \mid [B : A \cap B]$ , so  $[G : A] \leq [B : A \cap B]$ . We know  $[B : A \cap B] \leq [G : A]$  with equality iff  $G = AB$ , so  $G = AB$ , as desired.

P 9. If  $H$  is a proper subgroup of a finite group  $G$ , then  $\cup_{g \in G} gHg^{-1}$  is not the whole of  $G$ .

There are  $[G : N(H; G)]$   $H$ -conjugates in  $G$ . These groups are not disjoint because they all contain  $e$ , so

$$|\cup_{g \in G} gHg^{-1}| \leq [G : N(H; G)]|H| = [G] \cdot \frac{|H|}{|N(H; G)|}.$$

$N(H; G)$  contains  $H$ , so the set union contains at most  $|G|$  elements. Equality cannot hold since the union is not over disjoint sets, showing the result.

P 10. P 11. Let  $G$  be a group in which each proper subgroup is contained in a maximal subgroup of finite index in  $G$ . If every two maximal subgroups of  $G$  are conjugate in  $G$ , then  $G$  is a cyclic group.

By P 10, there exists an element  $x \in G$  not contained in

$$\cup_{g \in G} gHg^{-1},$$

where  $H$  is some fixed maximal subgroup. Consider the group  $\langle x \rangle$ , which cannot be proper. That is,  $G = \langle x \rangle$ , AWD.

P 12. Let  $S$  be a finite normal subset of a group  $G$  such that  $s^n = e$  for all  $s \in S$ . Show that every element of  $\langle S \rangle$  can be written as a product of at most  $(n-1)|S|$  elements of  $S$ , and that in particular  $\langle S \rangle$  is finite.

Let  $S = \{s_1, s_2, s_3, \dots, s_N\}$  and  $x = x_1^\pm x_2^\pm x_3^\pm \cdots x_k^\pm \in \langle S \rangle$ . Since  $s^{-1} = s^{n-1}$ , we may write  $x = y_1^{n_1} y_2^{n_2} y_3^{n_3} \cdots y_j^{n_j}$  for some positive integers  $n_i$  and  $y_i \in S$ .

Since  $S$  is normal in  $G$ , we have for all  $s, t \in S$ ,  $st = ts'$  for some  $s' \in S$ . Hence, we can move all copies of  $s_1$  to the front of  $x$ . There is the possibility that  $s' = s_1$ , so that additional copies of  $s_1$  are created when one copy of  $s_1$  is moved. However, since the number of terms being multiplied is constant, this process must eventually terminate. We can do this process inductively, moving all copies of  $s_2$  to the second position and so on.

We have  $x = s_1^{a_1} s_2^{a_2} s_3^{a_3} \cdots s_N^{a_N}$  for some  $a_i \in \mathbb{Z}^{\geq 0}$ . Since each  $s$  has order at most  $n$ , we can take each  $a_i \in [0, n-1]$ . If  $a_i$  is zero, we can ignore  $s_i^{a_i}$ . Hence,  $x$  is the product of at most  $(n-1)N$  terms of  $S$  upon expanding the exponents.

P 13. If  $H$  is a Hall subgroup of a finite group  $G$  and  $K$  is another subgroup of the same order, then  $N(H; K) = H \cap K$ .

P 14. If  $N$  is a normal subset of a group  $G$ , then its centralizer  $C(N; G)$  is a normal subgroup of  $G$ . If  $N$  is a characteristic subset of  $G$ , then  $C(N; G)$  is a characteristic subgroup of  $G$ .

Let  $c \in C(N; G)$ . Note

$$gcg^{-1}Ngc^{-1}g^{-1} = gcNc^{-1}g^{-1} = gNg^{-1} = N,$$

so  $C(N; G) \triangleleft G$ . Let  $\varphi \in \text{Aut}(G)$  and  $n \in N$ .  $\varphi(c)\varphi(n) = \varphi(n)\varphi(c)$ .  $n$  can be chosen so all of  $N$  is covered, so  $\varphi(c) \in C(N; G)$ .

P 15. Let  $N$  be a subgroup of  $Z(G)$ . Then  $N$  is a normal subgroup of  $G$ . If  $G/N$  is cyclic, then  $G$  is Abelian.

Let  $g \in G$ . Then  $gNg^{-1} = gg^{-1}N = N$ , so  $N \triangleleft G$ . Let  $a = g^x n$  and  $b = g^y m$ , where  $x, y \in \mathbb{Z}$  and  $n, m \in N$ . Then  $ab = g^{x+y} nm = ba$ , so  $G$  is Abelian.

P 17. Let  $N \triangleleft G$  be a normal subgroup of finite index and  $H < G$  be a subgroup of finite order such that  $[G : N] \perp |H|$ . Then  $H \subset N$ .

Let  $d = [G : N]$ ,  $k = |H|$  and  $h \in H$ .  $h^k = e \in N$  and  $h^d \in N$ . Since  $d \perp k$ , there exist  $x, y \in \mathbb{Z}$  such that  $1 = kx + dy$ , so  $h = h^{kx+dy} = (h^k)^x (h^d)^y \in N$ .

P 20. First let  $m_i = n/p_i^{e_i}$ . Then

$$\prod x_i = \prod x_i^{n/p_i^{e_i}} = x^{\sum n/p_i^{e_i}}.$$

Let  $N = \sum n/p_i^{e_i}$ . Note that  $\gcd(N, p^e) = \gcd(n/p^e, p^e) = 1$ , so  $\gcd(N, n) = 1$ . Pick  $M$  such that  $MN \equiv 1 \pmod{n}$ , so setting  $m_i = Mn/p_i^{e_i}$  gives our desired

$x_i$ s.

Now suppose we have  $y_1 y_2 \cdots y_s = x_1 x_2 \cdots x_s$  for  $y_i$  a  $p_i$  element. Since the  $y$ s commute and are distinct  $p$ -elements, the order of the LHS is the product of the orders of the  $y$ s. Similarly for the  $x$ s, so  $|x_i| = |y_i| = n/p_i^{e_i}$ . Exponentiate both sides by  $n/p^e$ , so  $x^{n/p^e} = y^{n/p^e}$ . Since  $n/p^e \perp p^e = |x| = |y|$ , we have  $x = y$ .